

FETALUS-NET: A HYBRID MAMBA-ATTENTION NETWORK FOR AUTOMATED RECOGNITION OF STANDARD FETAL ULTRASOUND PLANES

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ABSTRACT

The classification of fetal ultrasound (US) planes is a critical step in prenatal diagnosis and biometric assessment. Manual screening is time-consuming and subject to considerable inter-operator variability, hindering standardized evaluation. Existing automated methods often struggle to simultaneously capture fine-grained local features and long-range dependencies. To overcome this limitation, we propose FetalUS-Net, a four-stage hierarchical network designed for fetal US plane classification. Specifically, we introduce a novel dual-branch block named CBAMamba, which integrates a Convolutional Block Attention Module (CBAM) for local discriminative features with Mamba to capture long-range dependencies. We evaluated FetalUS-Net on two fetal US datasets collected from different regions. Experimental results demonstrate that our method achieves robust classification accuracy and strong cross-domain generalization, highlighting its potential as a reliable solution for automated fetal US plane recognition.

Index Terms— Fetal ultrasound plane classification, Biomedical image analysis, State space models

1. INTRODUCTION

Ultrasound (US) imaging is the preferred modality for prenatal evaluation owing to its non-invasive nature and absence of adverse effects on fetal health [1]. In clinical practice, key biometric parameters, such as fetal weight and head circumference [2], are measured from standard fetal anatomical planes. These measurements are essential for assessing fetal growth and development, facilitating the early detection and long-term monitoring of conditions like fetal growth restriction, structural malformations, and placental insufficiency [3]. Such evaluations provide critical evidence to guide perinatal management and predict long-term health outcomes. However, the subtle structural similarities between standard and non-standard planes (see Fig.1) pose considerable challenges [4]. Misidentification or acquisition of non-standard planes can increase measurement errors, reduce diagnostic accuracy, and compromise clinical decision-making [5]. Therefore, consistent acquisition of high-quality standard planes

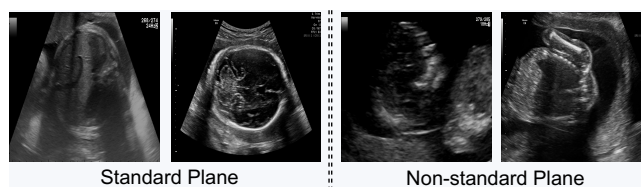


Fig. 1. Representative examples of standard (fetal abdomen and fetal brain) and non-standard fetal US planes.

is crucial for accurate maternal and fetal assessments. Nevertheless, manual plane selection by sonographers is time-consuming and subject to inter-operator variability [3, 6], which hinders efficiency and reproducibility in large-scale prenatal screening. Consequently, there is a pressing need for Computer-aided Diagnosis (CAD) [7] systems that can autonomously identify standard fetal planes, thereby accelerating image acquisition and improving diagnostic reliability.

In this paper, we propose FetalUS-Net, which is designed to assist sonographers in the rapid and accurate identification of standard fetal US planes. The core component of FetalUS-Net is the dual-branch CBAMamba block, which integrates the complementary strengths of local attention mechanisms and long-sequence modeling. Specifically, one branch incorporates Convolutional Block Attention Module (CBAM) [8] to emphasize local features, while the other leverages Mamba [9] to capture long-range contextual dependencies. Through the fusion of these two feature representations, FetalUS-Net captures complementary global and multi-scale local information, resulting in more discriminative feature representations and improved classification accuracy. We validated the effectiveness of FetalUS-Net on two independent datasets collected from distinct geographic regions. FetalUS-Net achieved not only high accuracy in fetal US plane classification but also strong cross-domain generalization, indicating its robustness and potential for clinical deployment.

The main contributions of this paper are summarized as follows: (1) We introduce the CBAMamba block, a novel dual-branch module that effectively integrates local and global feature representations. (2) We conduct extensive evaluations of FetalUS-Net on two multi-center, cross-regional

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fetal US datasets, verifying its robustness and generalizability. (3) We demonstrate the potential of FetalUS-Net for deployment in low-resource medical settings by adopting a pre-training and fine-tuning strategy, providing a practical pathway toward scalable computer-aided prenatal diagnosis.

2. RELATED WORKS

In recent years, artificial intelligence has become increasingly integral to fetal US image analysis, serving as a valuable tool to support clinical decision-making. Early research primarily focused on Convolutional Neural Networks (CNN) for fetal plane classification. Burgos-Artizzu et al. [1] evaluated 19 ImageNet-pretrained CNNs and reported that DenseNet-169 [10] achieved a 6.2% top-1 error, comparable to the 5.1% error rate of experienced sonographers. Building on this work, Prabakaran et al. [3] introduced the FPUS23 fetal phantom dataset, on which ResNet-34 [11] achieved nearly 99% accuracy in diagnostic-plane classification. Furthermore, FPUS23-pretrained ResNet-34 reached 91.92% accuracy on the Fetal-Planes-DB dataset after only one epoch of fine-tuning, whereas ImageNet-pretrained ResNet-34 required more than 16 epochs to reach similar performance. To address the challenges of low inter-class variance, Krishna et al. [12] proposed an adaptive channel weighting technique integrated into VGG-19 [13], which improved the automated identification of standard fetal planes. Despite these achievements, CNNs remain fundamentally limited by their local receptive fields, constraining their ability to capture long-range contextual dependencies.

To overcome these limitations, attention-based architectures, particularly Vision Transformers (ViT) [14], have been adopted for US image analysis. MedViTV2 [15], a hybrid Transformer that integrates local convolutional representations and global self-attention, has demonstrated both high accuracy and computational efficiency in fetal plane classification. Similarly, FetalMovNet, a hybrid CNN-attention network introduced in [16], was trained on 587 ten-frame sequences covering seven fetal anatomical structures and significantly outperformed existing baselines. Collectively, these findings highlight the advantages of hybrid CNN-Transformer frameworks in advancing automated prenatal imaging.

Recently, State Space Models (SSM) [9] have emerged as a powerful alternative to Transformers for sequence modeling. Although originally developed for natural language processing, Mamba has been successfully adapted to vision tasks, achieving superior performance in image analysis [17, 18]. MedMamba [19], a Mamba-based model for medical image classification, has shown competitive performance in fetal US, underscoring the potential of SSM for prenatal imaging. These advances suggest that incorporating Mamba into fetal US analysis offers a promising direction for balancing long-range dependency modeling with computational efficiency, which motivates the design of our proposed network.

3. FETALUS-NET

3.1. Model Architecture

In this section, we introduce FetalUS-Net in detail. As shown in Fig.2, FetalUS-Net is composed of four hierarchical stages, similar to prior works [17, 18, 19]. Specifically, Stage 1 is composed of several CBAMamba blocks, followed by a Patch Merging layer for downsampling. Stages 2 to 4 each contain multiple CBAMamba blocks and a Global Feature Perception (GFP) block [15], which performs downsampling while simultaneously enhancing global contextual representations.

Given an input image of size $H \times W \times C$, the stem layer uses four 3×3 convolutional blocks to generate overlapping patches of size $\frac{H}{4} \times \frac{W}{4} \times C_1$. Among these, the first and last blocks use a stride of 2 for spatial downsampling, while the two intermediate blocks adopt a stride of 1 to refine features without changing the resolution. Stage 1 applies Patch Merging layer, whereas Stages 2 to 4 apply GFP blocks to produce hierarchical feature maps of sizes $\frac{H}{8} \times \frac{W}{8} \times C_2$, $\frac{H}{16} \times \frac{W}{16} \times C_3$, and $\frac{H}{32} \times \frac{W}{32} \times C_4$, respectively. Finally, the classification results are generated using a global average pooling and a linear layer. The overall formulation of FetalUS-Net can be expressed as:

$$X_s = \text{Stem}(X) \quad (1)$$

$$X_d = \text{Downsample}(\text{CBAMamba}(X_s)) \quad (2)$$

$$\text{Output} = \text{Classifier}(\text{AvgPool}(X_d)) \quad (3)$$

3.2. CBAMamba Block

To effectively combine local attention with long-range dependency modeling, we designed the CBAMamba block as FetalUS-Net's core building unit. This block is inspired by the complementary strengths of the CBAM and the efficient sequence modeling capabilities of Mamba. Assuming an input $X \in \mathbb{R}^{H \times W \times C}$, the CBAMamba block splits X along the channel dimension into two parallel parts, $X_1, X_2 \in \mathbb{R}^{H \times W \times \frac{C}{2}}$, which are subsequently processed by the CBAM branch and the Mamba branch, respectively.

CBAM Branch. This branch applies the CBAM module, which sequentially refines features through channel attention and spatial attention [8]. Channel attention adaptively reweights feature maps to emphasize the most informative channels, while spatial attention highlights salient spatial locations. Consequently, the CBAM branch enhances local discriminative representations and improves the model's focus on relevant regions. This branch can be formulated as:

$$X_c = \sigma(\text{MLP}(\text{AP}(X_1)) + \text{MLP}(\text{MP}(X_1))) \otimes X_1 \quad (4)$$

$$X_{out1} = \sigma(\text{Conv}([\text{AP}(X_c); \text{MP}(X_c)])) \otimes X_c \quad (5)$$

where AP , MP , σ and Conv denote the average pooling, max pooling, sigmoid function, and convolution operation, respectively.

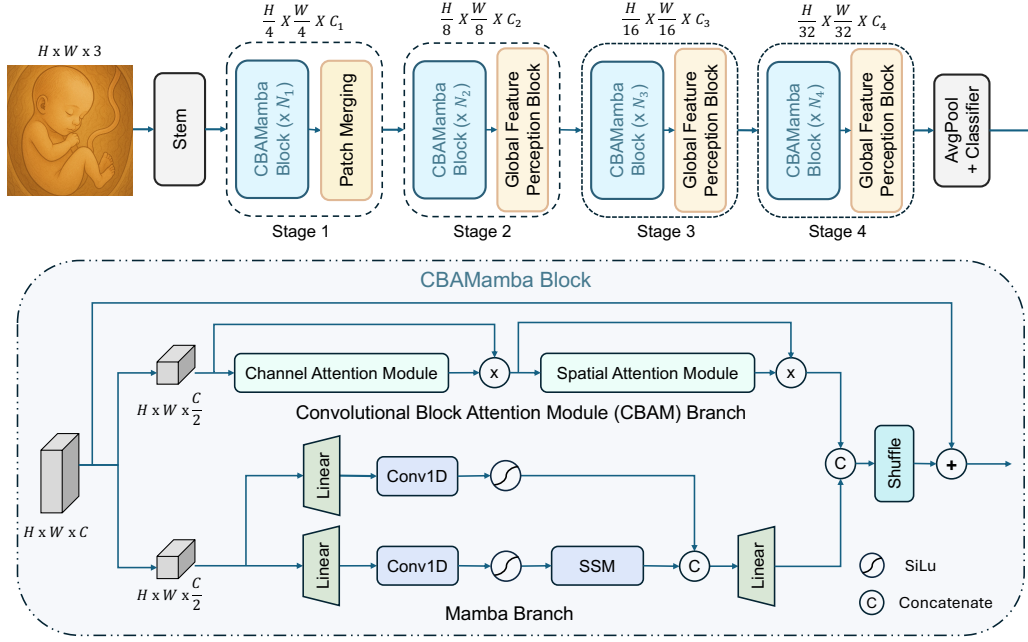


Fig. 2. Overall architecture of FetalUS-Net and structure of CBAMamba block. N_1 – N_4 indicate the numbers of CBAMamba blocks (2, 2, 8, 2), and C_1 – C_4 denote the embedding dimensions (128, 256, 512, 1024).

Mamba Branch. This branch processes X_2 with Mamba, where the selective SSM captures long-range dependencies across spatial dimensions. To compensate for the lost details caused by the sequence restrictions of SSM, we incorporated a parallel path composed of convolutional layers with SiLU activation, but without SSM. Finally, we concatenate the outputs of these two sub-branches and restore them to the original dimension through a linear layer. The output of the Mamba branch is as follows:

$$X_{b1} = SSM(SiLU(Conv(Linear(X_2)))) \quad (6)$$

$$X_{b2} = SiLU(Conv(Linear(X_2))) \quad (7)$$

$$X_{out2} = Linear(Concat(X_{b1}, X_{b2})) \quad (8)$$

where $Conv$ and SSM denote the convolution and selective scan operations, respectively.

After processing these two branches, we concatenate the results of these two branches. The concatenated outputs are further processed by a channel shuffle operation, which promotes cross-branch feature interaction and strengthens the integration of spatial and sequential representations. Finally, a residual connection is added to retain the original input, thereby generating the final output:

$$X_{out} = Shuffle(Concat(X_{out1}, X_{out2})) + X \quad (9)$$

This design allows CBAMamba to combine fine-grained local refinement with global sequential modeling. By stacking multiple CBAMamba blocks, the network progressively refines hierarchical features, achieving a robust balance between local detail representation and global contextual modeling.

4. EXPERIMENTS AND RESULTS

4.1. Implementation Details

The Fetal-Planes-DB dataset [1] contains 12,400 images acquired from BCNatal, a leading maternal-fetal medicine center in Spain. These images are categorized into six classes: fetal abdomen (711), fetal brain (3092), fetal femur (1040), fetal thorax (1718), maternal cervix (1626), and an other category (4213). In addition, the African dataset [20], collected from multiple healthcare institutions across Egypt, Algeria, Uganda, Ghana, and Malawi, comprises 450 images distributed among four fetal planes: fetal abdomen (125), fetal brain (125), fetal femur (125), and fetal thorax (75).

FetalUS-Net was trained for 300 epochs on the Fetal-Planes-DB dataset, which was randomly split into training, validation, and test sets with a ratio of 7:1.5:1.5. To evaluate the model's cross-domain adaptation, we fine-tuned the model, which was pretrained on the Fetal-Planes-DB dataset, for 50 epochs on the African dataset. Due to the limited data, we used a 10-fold cross-validation strategy. For both training phases, all input images were resized to $224 \times 224 \times 3$ pixels with a batch size of 64. The model was optimized using the AdamW optimizer with an initial learning rate of $1e-4$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and a weight decay of $1e-4$. The learning rate was dynamically adjusted using a cosine annealing schedule, and the network parameters were optimized using the cross-entropy loss function. We evaluated the model performance using macro-averaged metrics across all classes, including specificity, AUC, and accuracy.

Table 1. Classification results of comparison methods on the Fetal-Planes-DB dataset.

Model	Specificity(%) $\uparrow$	AUC $\uparrow$	Accuracy(%) $\uparrow$
ResNet-50[11]	98.9	0.992	95.0
Densenet-169[10]	98.9	0.993	95.4
ViT-B[14]	98.1	0.989	91.7
MedViT2-B[15]	99.0	N/A	95.3
VMamba-B[18]	98.7	0.994	93.8
MedMamba-B[19]	98.8	0.993	94.4
FetalUS-Net	99.0	0.995	95.9

Table 2. FetalUS-Net’s performance on the African dataset using 10-fold cross-validation.

Strategy	Specificity(%) $\uparrow$	AUC $\uparrow$	Accuracy(%) $\uparrow$
Non-pretrain	93.5	0.932	80.8
Pretrain	99.2	0.997	97.7

4.2. Experimental Results and Analysis

As shown in Table 1, FetalUS-Net outperforms existing CNN-based, Transformer-based, and Mamba-based methods on the Fetal-Planes-DB dataset in terms of AUC and accuracy. Furthermore, it achieves a Specificity score comparable to that of the best-performing MedViT2-B. These results demonstrate the effectiveness and robustness of FetalUS-Net for reliable fetal US image analysis.

The cross-domain evaluation on the African dataset is presented in Table 2. After fine-tuning, the pretrained model outperformed the non-pretrained model by 16.9% in accuracy. We observe that this substantial improvement can be attributed to the rich feature representations learned from large-scale data, which provide a strong prior for adaptation, even when the target domain differs in imaging equipment or patient populations. These findings highlight the practical utility of FetalUS-Net in low-resource clinical settings.

For deeper insight into the model’s performance, we present qualitative visualizations. Fig.3 illustrates heatmaps for each category in the Fetal-Planes-DB dataset, demonstrating that FetalUS-Net effectively highlights the most informative regions of interest. Fig.4 (top) presents the confusion matrix on the test set, which facilitates the analysis of class-wise prediction distributions. Furthermore, we utilize t-SNE feature embeddings to further evaluate the model’s classification performance. As shown in Fig.4 (bottom), the t-SNE embeddings exhibit clear intra-class clustering and inter-class separation among the standard fetal US plane categories.

5. CONCLUSION

In this paper, we proposed FetalUS-Net to effectively address the challenge of balancing local feature refinement and

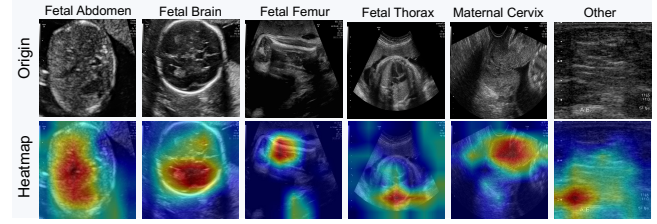


Fig. 3. The heatmap of the Fetal-Planes-DB dataset.

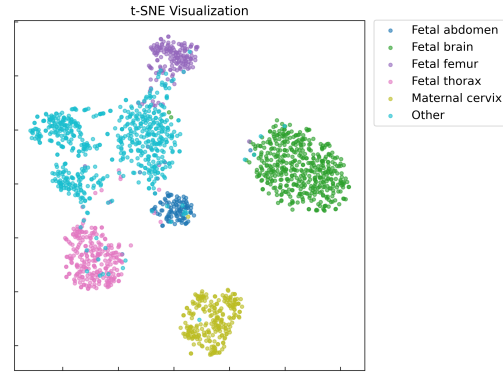
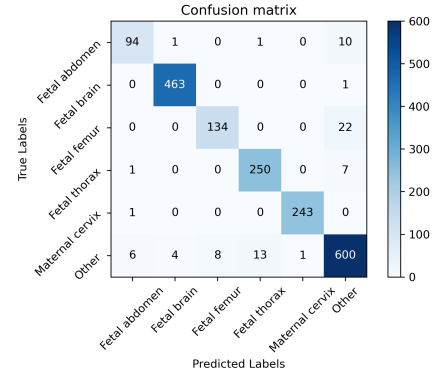


Fig. 4. Confusion matrix (top) and t-SNE visualization (bottom) of classification results on the Fetal-Planes-DB dataset.

global dependency modeling in fetal US plane classification. Through extensive experimentation on two independent datasets, we demonstrated that FetalUS-Net achieves competitive accuracy and exhibits strong cross-domain generalization. These findings underscore the potential of FetalUS-Net as a reliable tool for automated prenatal diagnosis and biometric assessment, especially its practical utility in low-resource clinical settings.

6. ACKNOWLEDGEMENTS

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7. REFERENCES

- [1] Xavier P Burgos-Artizzu, David Coronado-Gutiérrez, Valenzuela-Alcaraz, et al., “Evaluation of deep convolutional neural networks for automatic classification of common maternal fetal ultrasound planes,” *Scientific Reports*, vol. 10, no. 1, pp. 10200, 2020.
- [2] LJ Salomon, Z Alfirevic, V Berghella, CM Bilardo, GE Chalouhi, F Da Silva Costa, E Hernandez-Andrade, G Malinger, H Munoz, D Paladini, et al., “Isuog practice guidelines (updated): performance of the routine mid-trimester fetal ultrasound scan,” *Ultrasound in Obstetrics and Gynecology*, vol. 59, no. 6, pp. 840–856, 2022.
- [3] Bharath Srinivas Prabhakaran, Paul Hamelmann, Erik Ostrowski, and Muhammad Shafique, “Fpus23: an ultrasound fetus phantom dataset with deep neural network evaluations for fetus orientations, fetal planes, and anatomical features,” *IEEE Access*, vol. 11, pp. 58308–58317, 2023.
- [4] Maria Chiara Fiorentino, Francesca Pia Villani, Mariachiara Di Cosmo, Emanuele Frontoni, and Sara Moccia, “A review on deep-learning algorithms for fetal ultrasound-image analysis,” *Medical image analysis*, vol. 83, pp. 102629, 2023.
- [5] Baoping Zhu, Fan Yang, Hongliang Duan, and Zhipeng Gao, “Automatic detection of standard planes in fetal ultrasound images based on convolutional neural networks and ensemble learning,” *Current Bioinformatics*, vol. 20, no. 5, pp. 443–451, 2025.
- [6] Helena R Torres, Pedro Morais, Bruno Oliveira, Cahit Birdir, Mario Rüdiger, Jaime C Fonseca, and João L Vilaça, “A review of image processing methods for fetal head and brain analysis in ultrasound images,” *Computer methods and programs in biomedicine*, vol. 215, pp. 106629, 2022.
- [7] Li Liu, Haocheng Sun, and Fanzhang Li, “A lie group kernel learning method for medical image classification,” *Pattern Recognition*, vol. 142, pp. 109735, 2023.
- [8] Sanghyun Woo, Jongchan Park, Joon-Young Lee, and In So Kweon, “Cbam: Convolutional block attention module,” in *Proceedings of the European Conference on Computer Vision (ECCV)*, September 2018.
- [9] Albert Gu and Tri Dao, “Mamba: Linear-time sequence modeling with selective state spaces,” *arXiv preprint arXiv:2312.00752*, 2023.
- [10] Gao Huang, Zhuang Liu, Laurens Van Der Maaten, and Kilian Q Weinberger, “Densely connected convolutional networks,” in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2017, pp. 4700–4708.
- [11] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun, “Deep residual learning for image recognition,” in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770–778.
- [12] Thunakala Bala Krishna and Priyanka Kokil, “Integration of a deep convolutional neural network with adaptive channel weight technique for automated identification of standard fetal biometry planes,” *IEEE Transactions on Instrumentation and Measurement*, vol. 73, pp. 1–11, 2024.
- [13] Karen Simonyan and Andrew Zisserman, “Very deep convolutional networks for large-scale image recognition,” *arXiv preprint arXiv:1409.1556*, 2014.
- [14] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al., “An image is worth 16x16 words: Transformers for image recognition at scale,” *arXiv preprint arXiv:2010.11929*, 2020.
- [15] Omid Nejati Manzari, Hojat Asgariandehkordi, Taha Koleilat, Yiming Xiao, and Hassan Rivaz, “Medical image classification with kan-integrated transformers and dilated neighborhood attention,” *arXiv preprint arXiv:2502.13693*, 2025.
- [16] Musa Turkan, Emre Dandil, Furkan Erturk Urfali, and Mehmet Korkmaz, “Fetalmovnet: A novel deep learning model based on attention mechanism for fetal movement classification in us,” *IEEE Access*, 2025.
- [17] Ali Hatamizadeh and Jan Kautz, “Mambavision: A hybrid mamba-transformer vision backbone,” in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 25261–25270.
- [18] Yue Liu, Yunjie Tian, Yuzhong Zhao, Hongtian Yu, Lingxi Xie, Yaowei Wang, Qixiang Ye, Jianbin Jiao, and Yunfan Liu, “Vmamba: Visual state space model,” *Advances in neural information processing systems*, vol. 37, pp. 103031–103063, 2024.
- [19] Yubiao Yue and Zhenzhang Li, “Medmamba: Vision mamba for medical image classification,” *arXiv preprint arXiv:2403.03849*, 2024.
- [20] Carla Sendra-Balcells, Víctor M Campello, Torrents-Barrena, et al., “Generalisability of fetal ultrasound deep learning models to low-resource imaging settings in five african countries,” *Scientific reports*, vol. 13, no. 1, pp. 2728, 2023.